

Derivation of the Lense-Thirring Metric

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Context

The conceptual structure of this derivation follows material introduced in advanced coursework at the Universitat de Barcelona. All detailed calculations, formal development, and typesetting are the independent work of the author.

Introduction

In this section, we will calculate the correction to the Schwarzschild metric due to the rotation of the source. We will do this by considering non-relativistic rotational velocities and working in the weak-field limit.

We consider a spherical distribution of mass M and radius R , with uniform density ρ and angular velocity Ω . All these quantities are constant. We assume that the source rotates at velocities much lower than the speed of light, that is, $R\Omega \ll c$. Treating the source as a perfect fluid without pressure, the energy-momentum tensor is:

$$T^{\mu\nu} = (\rho + P)u^\mu u^\nu + Pg^{\mu\nu}$$

If the pressure is zero ($P = 0$), then:

$$T^{\mu\nu} = \rho u^\mu u^\nu$$

The four-velocity of the fluid, by normalization, is $u^\mu = \gamma(1, \mathbf{v})$, where γ is the Lorentz factor. For this system, $\|\mathbf{v}\| = \Omega R \ll c \implies \gamma \approx 1$, so that:

$$T^{00} = \rho u^0 u^0 = \rho c^2$$

$$T^{0i} = \rho u^0 u^i = \rho c v^i$$

$$T^{ij} = \rho u^i u^j = \rho v^i v^j \sim \mathcal{O}((R\Omega)^2)$$

The energy-momentum tensor at first order results in:

$$T^{\mu\nu} \simeq \begin{pmatrix} \rho c^2 & \rho c v^x & \rho c v^y & \rho c v^z \\ \rho c v^x & 0 & 0 & 0 \\ \rho c v^y & 0 & 0 & 0 \\ \rho c v^z & 0 & 0 & 0 \end{pmatrix} = \rho c \begin{pmatrix} c & v^x & v^y & v^z \\ v^x & 0 & 0 & 0 \\ v^y & 0 & 0 & 0 \\ v^z & 0 & 0 & 0 \end{pmatrix}$$

1. Linearized Field Equations

Starting from the linearized Einstein equations in the Lorenz gauge (Harmonic gauge), $g^{\mu\nu}\Gamma_{\mu\nu}^\alpha = 0$, which can also be expressed as $\partial^\mu \bar{h}_{\mu\nu} = 0$ in terms of the trace-reversed tensor

$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$. With $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ and $|h| \ll 1$, we can switch to the purely contravariant form. Under the weak-field approximation, we will use only $\eta_{\mu\nu}$, so that:

$$\square \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu} \implies \square \bar{h}^{\mu\nu} = -\frac{16\pi G}{c^4} T^{\mu\nu}$$

The Einstein equations to solve in this case are:

$$\begin{aligned} \square \bar{h}^{00} &= -\frac{16\pi G}{c^4} T^{00} = -\frac{16\pi G}{c^4} \rho c^2 = -\frac{16\pi G}{c^2} \rho \\ \square \bar{h}^{0i} &= -\frac{16\pi G}{c^4} T^{0i} = -\frac{16\pi G}{c^4} \rho c v^i = -\frac{16\pi G}{c^3} \rho v^i \\ \square \bar{h}^{ij} &= -\frac{16\pi G}{c^4} T^{ij} \approx 0 \end{aligned}$$

Assuming the body rotates around the z -axis, we only need to use the 3-velocity $\mathbf{v} = \boldsymbol{\Omega} \times \mathbf{r}$ in Cartesian coordinates:

$$\mathbf{v} = \boldsymbol{\Omega} \times \mathbf{r} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & \Omega \\ x & y & z \end{vmatrix} = -\mathbf{i}(\Omega y) + \mathbf{j}(\Omega x) = \Omega(-y\mathbf{i} + x\mathbf{j})$$

2. Metric Perturbation at Large Distances

In the static limit, $\square = \nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}$, so that:

$$\nabla^2 \bar{h}^{\mu\nu} = -\frac{16\pi G}{c^4} T^{\mu\nu}$$

Using the general solution to Poisson's equation $\nabla^2 f = g$:

$$\bar{h}^{\mu\nu}(\mathbf{x}) = \frac{4G}{c^4} \int \frac{T^{\mu\nu}(\mathbf{y})}{\|\mathbf{x} - \mathbf{y}\|} d^3y$$

If we assume large distances from the source ($r \gg R$), we can initially try the approximation: $\|\mathbf{x} - \mathbf{y}\| \simeq \|\mathbf{x}\|$. Defining $r := \|\mathbf{x}\|$ and $r' := \|\mathbf{y}\|$:

$$\begin{aligned} \bar{h}^{00}(\mathbf{x}) &= \frac{4G}{c^4 r} \int T^{00}(\mathbf{y}) d^3y = \frac{4G}{c^4 r} \int \rho c^2 d^3y = \frac{4G}{c^2 r} \int \rho d^3y = \frac{4GM}{c^2 r} \\ \bar{h}^{0x}(\mathbf{x}) &= \frac{4G}{c^4 r} \int T^{0i}(\mathbf{y}) d^3y = \frac{4G}{c^4 r} \int \rho c v^x d^3y = -\frac{4G\Omega\rho}{c^3 r} \int y' d^3y \\ &= -\frac{4G\Omega\rho}{c^3 r} \int_0^R \int_0^\pi \int_0^{2\pi} r' \sin\theta \cos\phi r'^2 \sin\theta dr' d\theta d\phi \\ &= -\frac{4G\Omega\rho}{c^3 r} \int_0^R (r')^3 dr' \int_0^\pi \sin^2\theta d\theta \int_0^{2\pi} \cos\phi d\phi \\ &= 0 \end{aligned}$$

Therefore, we must approximate to a higher order if we want a non-zero contribution. We expand $\|\mathbf{x} - \mathbf{y}\|$ to first order:

$$\|\mathbf{x} - \mathbf{y}\| = \sqrt{(\mathbf{x} - \mathbf{y}) \cdot (\mathbf{x} - \mathbf{y})} = \sqrt{r^2 - 2\mathbf{x} \cdot \mathbf{y} + (r')^2} \simeq \sqrt{r^2 - 2\mathbf{x} \cdot \mathbf{y}} = r \sqrt{1 - 2 \frac{\mathbf{x} \cdot \mathbf{y}}{r^2}}$$

$$\simeq r \left(1 - \frac{\mathbf{x} \cdot \mathbf{y}}{r^2} \right)$$

$$\Rightarrow \frac{1}{\|\mathbf{x} - \mathbf{y}\|} \simeq \frac{1}{r} \left(1 + \frac{\mathbf{x} \cdot \mathbf{y}}{r^2} \right)$$

Since we have already checked that the contribution from the constant term is zero, we can write:

$$\begin{aligned} \bar{h}^{0x}(\mathbf{x}) &\simeq \frac{4G}{c^4 r^3} \int \rho c v^x(\mathbf{x} \cdot \mathbf{y}) d^3 y = -\frac{4G\Omega\rho}{c^3 r^3} \int y'(xx' + yy' + zz') d^3 y \\ &= -\frac{4G\Omega\rho}{c^3 r^3} \left[x \underbrace{\int (y'x') d^3 y}_{I_1} + y \underbrace{\int (y'y') d^3 y}_{I_2} + z \underbrace{\int (y'z') d^3 y}_{I_3} \right] \\ \bar{h}^{0y}(\mathbf{x}) &\simeq \frac{4G}{c^4 r^3} \int \rho c v^y(\mathbf{x} \cdot \mathbf{y}) d^3 y = \frac{4G\Omega\rho}{c^3 r^3} \int x'(xx' + yy' + zz') d^3 y \\ &= \frac{4G\Omega\rho}{c^3 r^3} \left[x \underbrace{\int (x'x') d^3 y}_{I_4} + y \underbrace{\int (x'y') d^3 y}_{I_1} + z \underbrace{\int (x'z') d^3 y}_{I_5} \right] \end{aligned}$$

Let's investigate the integrals separately:

$$\begin{aligned} I_1 &= \int_0^R \int_0^\pi \int_0^{2\pi} r' \sin \theta \cos \phi r' \sin \theta \sin \phi r'^2 \sin \theta dr' d\theta d\phi \\ &= \int_0^R r'^4 dr' \int_0^\pi \sin^3 \theta d\theta \int_0^{2\pi} \cancel{\cos \phi \sin \phi} d\phi \\ &= 0 \iff \int_0^{2\pi} \cos \phi \sin \phi d\phi = \left[\frac{1}{2} \sin^2 \phi \right]_0^{2\pi} = 0 \\ I_2 &= \int_0^R \int_0^\pi \int_0^{2\pi} r'^2 \sin^2 \theta \sin^2 \phi r'^2 \sin \theta dr' d\theta d\phi = \int_0^R r'^4 dr' \int_0^\pi \sin^3 \theta d\theta \int_0^{2\pi} \sin^2 \phi d\phi \\ &= \frac{R^5}{5} \int_0^\pi (1 - \cos^2 \theta) \sin \theta d\theta \int_0^{2\pi} \frac{1}{2} (1 - \cos(2\phi)) d\phi \\ &= \frac{R^5}{5} \int_0^\pi - (1 - \cos^2 \theta) d(\cos \theta) \left[\frac{\phi}{2} - \frac{1}{4} \sin(2\phi) \right]_0^{2\pi} = \frac{\pi R^5}{5} \left[\cos \theta - \frac{1}{3} \cos^3 \theta \right]_0^\pi \\ &= \frac{\pi R^5}{5} \left(\left[1 - \frac{1}{3} \right] - \left[-1 + \frac{1}{3} \right] \right) = \frac{\pi R^5}{5} \left(2 - \frac{2}{3} \right) = \frac{4\pi R^5}{15} \\ I_3 &= \int_0^R \int_0^\pi \int_0^{2\pi} r' \sin \theta \sin \phi r' \cos \theta r'^2 \sin \theta dr' d\theta d\phi \\ &= \int_0^R r'^4 dr' \int_0^\pi \sin^2 \theta \cos \theta d\theta \int_0^{2\pi} \cancel{\sin \phi} d\phi = 0 \iff \int_0^{2\pi} \sin \phi d\phi = [-\cos \phi]_0^{2\pi} = 0 \\ I_4 &= \int_0^R \int_0^\pi \int_0^{2\pi} r'^2 \sin^2 \theta \cos^2 \phi r'^2 \sin \theta dr' d\theta d\phi = \int_0^R r'^4 dr' \int_0^\pi \sin^3 \theta d\theta \int_0^{2\pi} \cos^2 \phi d\phi \\ &= I_2 \iff \int_0^{2\pi} \cos^2 \phi d\phi = 2\pi - \int_0^{2\pi} \sin^2 \phi d\phi = \pi \end{aligned}$$

$$\begin{aligned}
I_5 &= \int_0^R \int_0^\pi \int_0^{2\pi} r' \sin \theta \cos \phi r'^2 \sin \theta dr' d\theta d\phi \\
&= \int_0^R r'^4 dr' \int_0^\pi \sin^2 \theta \cos \theta d\theta \int_0^{2\pi} \cancel{\cos \phi} d\phi = 0 \iff \int_0^{2\pi} \cos \phi d\phi = [\sin \phi]_0^{2\pi} = 0
\end{aligned}$$

Substituting the results of the integrals:

$$\begin{aligned}
\bar{h}^{0x}(\mathbf{x}) &= -\frac{4G\Omega\rho}{c^3r^3} y \frac{4\pi R^5}{15} = -\frac{4G\Omega y}{c^3r^3} \frac{R^2}{5} \rho \left(\frac{4}{3}\pi R^3 \right) = -\frac{4G\Omega M R^2}{5c^3r^3} y \\
\bar{h}^{0y}(\mathbf{x}) &= \frac{4G\Omega\rho}{c^3r^3} x \frac{4\pi R^5}{15} = \frac{4G\Omega x}{c^3r^3} \frac{R^2}{5} \rho \left(\frac{4}{3}\pi R^3 \right) = \frac{4G\Omega M R^2}{5c^3r^3} x
\end{aligned}$$

3. The Line Element in Cartesian and Spherical Coordinates

To write the line element in any coordinate system, we use the metric. In the linearized case and in Cartesian coordinates:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = (\eta_{\mu\nu} + h_{\mu\nu}) dx^\mu dx^\nu$$

So the objective is to find $h_{\mu\nu}$. By definition of $\bar{h}_{\mu\nu}$:

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h \implies h_{\mu\nu} = \bar{h}_{\mu\nu} + \frac{1}{2}\eta_{\mu\nu}h$$

By definition of the trace of $\bar{h}_{\mu\nu}$:

$$\bar{h} = \eta^{\mu\nu}\bar{h}_{\mu\nu} = \eta^{\mu\nu} \left(h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h \right) = h - \frac{1}{2}(4)h = -h$$

So that we can write:

$$h_{\mu\nu} = \bar{h}_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\bar{h}$$

As in this case the only non-zero diagonal component of $\bar{h}_{\mu\nu}$ is $\bar{h}_{00}$, we have:

$$\bar{h} = \eta_{00}\bar{h}^{00} = -\bar{h}_{00} = \eta^{00}\bar{h}_{00} = -\bar{h}^{00} = -\frac{4GM}{c^2r} \implies h_{\mu\nu} = \bar{h}_{\mu\nu} + \frac{2GM}{c^2r}\eta_{\mu\nu}$$

Evaluating the components of the tensor $h_{\mu\nu}$:

$$\begin{aligned}
h_{00} &= \bar{h}_{00} - \frac{2GM}{rc^2} = \frac{4GM}{rc^2} - \frac{2GM}{rc^2} = \frac{2GM}{rc^2} \\
h_{ij} &= \cancel{\bar{h}_{ij}} - \frac{1}{2}\eta_{ij} \left(-\frac{4GM}{rc^2} \right) = \frac{2GM}{rc^2}\delta_{ij} \\
h_{0i} &= \bar{h}_{0i} - \frac{1}{2}\eta_{0i} \frac{2GM}{rc^2} = \bar{h}_{0i} = \eta_{00}\eta_{ij}\bar{h}^{0j} = -\bar{h}^{0i} \\
\implies h_{0x} &= -\bar{h}^{0x} = \frac{4G\Omega M R^2}{5c^3r^3} y \quad h_{0y} = -\bar{h}^{0y} = -\frac{4G\Omega M R^2}{5c^3r^3} x
\end{aligned}$$

Now we can write the line element using the expression given previously:

$$ds^2 = (\eta_{00} + h_{00}) c^2 dt^2 + (\eta_{ij} + h_{ij}) dx^i dx^j + 2(\eta_{0i} + h_{0i}) dx^0 dx^i$$

$$\begin{aligned}
&= \left(-1 + \frac{2GM}{rc^2}\right) c^2 dt^2 + \left(\delta_{ij} + \delta_{ij} \frac{2GM}{rc^2}\right) dx^i dx^j + \frac{8GMR^2\Omega}{5r^3c^2} cdt (ydx - xdy) \\
&= -\left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 + \left(1 + \frac{2GM}{rc^2}\right) \delta_{ij} dx^i dx^j + \frac{8GMR^2\Omega}{5r^3c^2} (ydx - xdy) dt
\end{aligned}$$

Knowing how the Euclidean metric transforms into spherical coordinates, we can directly write:

$$\delta_{ij} dx^i dx^j = dx^2 + dy^2 + dz^2 = dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2)$$

For the case of $ydx - xdy$, we know they generate a rotation around the z -axis so that:

$$ydx - xdy = -r^2 \sin^2 \theta d\phi$$

Since $r \sin \theta$ represents the distance to the rotation axis. But, we can also do it explicitly:

$$\begin{aligned}
ydx &= r \sin \theta \sin \phi (\sin \theta \cos \phi dr + r \cos \phi \cos \theta d\theta - r \sin \theta \sin \phi d\phi) \\
&= r \sin^2 \theta \cos \phi \sin \phi dr + r^2 \sin \theta \cos \theta \sin \phi \cos \phi d\theta - r^2 \sin^2 \theta \sin^2 \phi d\phi \\
xdy &= r \sin \theta \cos \phi (\sin \theta \sin \phi dr + r \cos \phi \cos \theta d\theta + r \sin \theta \cos \phi d\phi) \\
&= r \sin^2 \theta \cos \phi \sin \phi dr + r^2 \sin \theta \cos \theta \sin \phi \cos \phi d\theta + r^2 \sin^2 \theta \cos^2 \phi d\phi \\
\implies ydx - xdy &= -r^2 \sin^2 \theta (\sin^2 \phi + \cos^2 \phi) d\phi = -r^2 \sin^2 \theta d\phi
\end{aligned}$$

Substituting into the line element found previously:

$$ds^2 = -\left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 + \left(1 + \frac{2GM}{rc^2}\right) [dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2)] - \frac{8GMR^2\Omega}{5rc^2} \sin^2 \theta d\phi dt$$

4. Isotropic to Standard Coordinate Transformation

With the proposed substitution $\tilde{r} = r + \frac{GM}{c^2} \implies r = \tilde{r} - \frac{GM}{c^2}$, where we discard higher order terms. Note also that $d\tilde{r} = dr$. Now we must calculate the behavior of the term that includes the spatial line element:

$$\begin{aligned}
I &:= \left(1 + \frac{2GM}{rc^2}\right) [dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2)] \\
&\simeq \left(1 + \frac{2GM}{\tilde{r}c^2}\right) \left[d\tilde{r}^2 + \left(\tilde{r} - \frac{GM}{c^2}\right)^2 (d\theta^2 + \sin^2 \theta d\phi^2)\right] \\
&= \left(1 + \frac{2GM}{\tilde{r}c^2}\right) d\tilde{r}^2 + \left(1 + \frac{2GM}{\tilde{r}c^2}\right) \left(\tilde{r}^2 - \frac{2GM\tilde{r}}{c^2} + \frac{G^2M^2}{c^4}\right) (d\theta^2 + \sin^2 \theta d\phi^2) \\
&\simeq \left(1 + \frac{2GM}{\tilde{r}c^2}\right) d\tilde{r}^2 + \tilde{r}^2 \left(1 + \frac{2GM}{\tilde{r}c^2}\right) \left(1 - \frac{2GM}{\tilde{r}c^2}\right) (d\theta^2 + \sin^2 \theta d\phi^2) \\
&\simeq \left(1 + \frac{2GM}{\tilde{r}c^2}\right) d\tilde{r}^2 + \tilde{r}^2 \left(1 - \frac{4G^2M^2}{\tilde{r}^2c^4}\right) (d\theta^2 + \sin^2 \theta d\phi^2) \\
&\simeq \left(1 + \frac{2GM}{\tilde{r}c^2}\right) d\tilde{r}^2 + \tilde{r}^2 (d\theta^2 + \sin^2 \theta d\phi^2)
\end{aligned}$$

The Lense-Thirring Metric

With this, the line element becomes the Lense-Thirring metric, the precursor to the Kerr metric:

$$ds^2 = - \left(1 - \frac{2GM}{\tilde{r}c^2}\right) c^2 dt^2 + \left(1 + \frac{2GM}{\tilde{r}c^2}\right) d\tilde{r}^2 + \tilde{r}^2 \left(d\theta^2 + \sin^2 \theta d\phi^2\right) - \frac{8GM}{5\tilde{r}c^2} \Omega R^2 \sin^2 \theta dt d\phi$$

5. Christoffel Symbols of the Lense-Thirring Geometry

With the definition of the Christoffel symbols:

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu})$$

Identifying the components of the metric:

$$g_{00} = - \left(1 - \frac{2GM}{\tilde{r}c^2}\right), \quad g_{\tilde{r}\tilde{r}} = \left(1 + \frac{2GM}{\tilde{r}c^2}\right), \quad g_{\theta\theta} = \tilde{r}^2, \quad g_{\phi\phi} = \tilde{r}^2 \sin^2 \theta, \quad g_{0\phi} = - \frac{2GMa}{\tilde{r}c^3} \sin^2 \theta$$

Where $a \equiv \frac{2}{5}\Omega R^2$ is the angular momentum per unit mass. The requested Christoffel symbols can be calculated taking into account that the problem is stationary ($\partial_0 f = 0$) and there is symmetry around the z -axis ($\partial_\phi f = 0$).

$$\begin{aligned} \Gamma_{00}^0 &= \frac{1}{2} g^{0\sigma} (\partial_0 g_{0\sigma} + \partial_0 g_{0\sigma} - \partial_\sigma g_{00}) = -\frac{1}{2} g^{0\sigma} \partial_\sigma g_{00} = -\frac{1}{2} g^{00} \partial_0 g_{00} - \frac{1}{2} g^{0\phi} \partial_\phi g_{00} = 0 \\ \Gamma_{\phi\phi}^0 &= \frac{1}{2} g^{0\sigma} (\partial_\phi g_{\phi\sigma} + \partial_\phi g_{\phi\sigma} - \partial_\sigma g_{\phi\phi}) = -\frac{1}{2} g^{0\sigma} \partial_\sigma g_{\phi\phi} = -\frac{1}{2} g^{00} \partial_0 g_{\phi\phi} - \frac{1}{2} g^{0\phi} \partial_\phi g_{\phi\phi} = 0 \\ \Gamma_{0\phi}^0 &= \frac{1}{2} g^{0\sigma} (\partial_0 g_{\phi\sigma} + \partial_\phi g_{0\sigma} - \partial_\sigma g_{0\phi}) = -\frac{1}{2} g^{0\sigma} \partial_\sigma g_{0\phi} = -\frac{1}{2} g^{00} \partial_0 g_{0\phi} - \frac{1}{2} g^{0\phi} \partial_\phi g_{0\phi} = 0 \\ \Gamma_{00}^\phi &= \frac{1}{2} g^{\phi\sigma} (\partial_0 g_{0\sigma} + \partial_0 g_{0\sigma} - \partial_\sigma g_{00}) = -\frac{1}{2} g^{\phi\sigma} \partial_\sigma g_{00} = -\frac{1}{2} g^{\phi 0} \partial_0 g_{00} - \frac{1}{2} g^{\phi\phi} \partial_\phi g_{00} = 0 \\ \Gamma_{\phi\phi}^\phi &= \frac{1}{2} g^{\phi\sigma} (\partial_\phi g_{\phi\sigma} + \partial_\phi g_{\phi\sigma} - \partial_\sigma g_{\phi\phi}) = -\frac{1}{2} g^{\phi\sigma} \partial_\sigma g_{\phi\phi} = -\frac{1}{2} g^{\phi 0} \partial_0 g_{\phi\phi} - \frac{1}{2} g^{\phi\phi} \partial_\phi g_{\phi\phi} = 0 \\ \Gamma_{0\phi}^\phi &= \frac{1}{2} g^{\phi\sigma} (\partial_0 g_{\phi\sigma} + \partial_\phi g_{0\sigma} - \partial_\sigma g_{0\phi}) = -\frac{1}{2} g^{\phi\sigma} \partial_\sigma g_{0\phi} = -\frac{1}{2} g^{\phi 0} \partial_0 g_{0\phi} - \frac{1}{2} g^{\phi\phi} \partial_\phi g_{0\phi} = 0 \end{aligned}$$

For the non-zero symbols, we must first calculate the inverse metric. Since the off-diagonal terms are $g_{0\phi} \sim \mathcal{O}(c^{-3})$, we treat the metric as diagonal in order to find the inverse at the desired order. In this way, the only term that interests us now is $g^{\tilde{r}\tilde{r}} \simeq \frac{1}{g_{\tilde{r}\tilde{r}}}$:

$$g^{\tilde{r}\tilde{r}} \simeq \frac{1}{1 + \frac{2GM}{\tilde{r}c^2}} \simeq 1 - \frac{2GM}{\tilde{r}c^2}$$

Then, the Christoffel symbols can be calculated explicitly:

$$\begin{aligned} \Gamma_{00}^{\tilde{r}} &= \frac{1}{2} g^{\tilde{r}\sigma} (\partial_0 g_{0\sigma} + \partial_0 g_{0\sigma} - \partial_\sigma g_{00}) = -\frac{1}{2} g^{\tilde{r}\sigma} \partial_\sigma g_{00} = -\frac{1}{2} g^{\tilde{r}\tilde{r}} \partial_{\tilde{r}} g_{00} \\ &= -\frac{1}{2} \left(1 - \frac{2GM}{\tilde{r}c^2}\right) \left[-\frac{2GM}{\tilde{r}^2 c^2}\right] = \frac{GM}{\tilde{r}^2 c^2} \left(1 - \frac{2GM}{\tilde{r}c^2}\right) \end{aligned}$$

$$\begin{aligned}
\Gamma_{\phi\phi}^{\tilde{r}} &= \frac{1}{2} g^{\tilde{r}\sigma} \left(\cancel{\partial_\phi g_{\phi\sigma}} + \cancel{\partial_\phi g_{\phi\sigma}} - \partial_\sigma g_{\phi\phi} \right) = -\frac{1}{2} g^{\tilde{r}\sigma} \partial_\sigma g_{\phi\phi} = -\frac{1}{2} g^{\tilde{r}\tilde{r}} \partial_{\tilde{r}} g_{\phi\phi} \\
&= -\frac{1}{2} \left(1 - \frac{2GM}{\tilde{r}c^2} \right) [2\tilde{r} \sin^2 \theta] = -\left(1 - \frac{2GM}{\tilde{r}c^2} \right) \tilde{r} \sin^2 \theta \\
\Gamma_{0\phi}^{\tilde{r}} &= \frac{1}{2} g^{\tilde{r}\sigma} \left(\cancel{\partial_0 g_{\phi\sigma}} + \cancel{\partial_\phi g_{0\sigma}} - \partial_\sigma g_{0\phi} \right) = -\frac{1}{2} g^{\tilde{r}\sigma} \partial_\sigma g_{0\phi} = -\frac{1}{2} g^{\tilde{r}\tilde{r}} \partial_{\tilde{r}} g_{0\phi} \\
&= -\frac{1}{2} \left(1 - \frac{2GM}{\tilde{r}c^2} \right) \left[-\frac{2GMa}{\tilde{r}^2 c^3} \sin^2 \theta \right] = -\left(1 - \frac{2GM}{\tilde{r}c^2} \right) \frac{GMa}{\tilde{r}^2 c^3} \sin^2 \theta
\end{aligned}$$